

Question	Answer	Marks
8(a)	four diodes shown (one in each gap) with correct circuit symbols	B1
	all four diodes connected in correct direction (pointing left to right)	B1
8(b)(i)	three lines showing exponential decay from the peaks at $0.5T$, $1.0T$ and $1.5T$, ending at the point where the decay meets the next rising peak	B1
	three curved lines rising along the upwards dotted lines from the minima to the peaks at $1.0T$, $1.5T$ and $2.0T$	B1
	minimum V at $(2/3) V_0$	B1
8(b)(ii)	from graph: discharge time (from V_0 to $(2/3)V_0$) = $(11/30) T$	C1
	$(2/3) V_0 = V_0 \exp(- (11/30)T/\tau)$	C1
	$\ln(2/3) = -11T/30\tau$	A1
	$\tau = 0.90 T$	
8(b)(iii)	time constant is increased	B1
	line shows a larger minimum voltage or smaller difference between maximum and minimum voltages	B1

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